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Abstract

Little work has accounted for congestion, using data that reflects on drive pattern, traffic volume, and speed - approach that can provide snapshot of congestion, to examine the association between traffic emissions and human health. In this study, we performed a health risk assessment of PM_{2.5} emissions during congestion periods in the Greater Toronto and Hamilton Area (GTHA), Canada. Specifically, we used a micro-level approach that combines Stochastic User Equilibrium Traffic Assignment Algorithm and MOVES emission model to estimate emissions considering congestion conditions, subsequently we applied a concentration-response function to estimate PM_{2.5}-related mortality, and finally we estimated the associated health costs. Our results suggest that traffic congestion has a substantial impact on human health and economy in the GTHA, especially at the most congested period (7:00 am). Considering daily mortality, our results showed an impact of 206 (boundary test 95%: 116; 297) and 119 (boundary test 95%: 67; 171) deaths per year (all-cause and cardiovascular mortality, respectively). The economic impact from daily mortality is approximately \$1.3 billion (boundary test 95%: 0.8; 1.9), and \$778 million (boundary test 95%: 478; 981), for all-cause and cardiovascular mortality, respectively. Our study can guide reliable projections of transportation and air pollution levels, improving the capability of the medical community to prepare for future trends.

Keywords

Traffic congestion, air pollution, health risk assessment, public health.

1. INTRODUCTION

The effects of traffic congestion have long been a concern in most metropolitan areas. By increasing travel times, traffic congestion increases the costs of travel and reduces accessibility (Weber and Kwan, 2002). Exposure to such congestion has also been associated with other negative effects, including increases in driver stress (Higgins et al., 2017), increases in feelings of time pressure and decreases in an individual's reported subjective wellbeing (Hilbrecht et al., 2014), and a slowing of metropolitan economic growth (Sweet, 2014).

From an environmental perspective, traffic congestion also results in increases in fuel consumption (Bigazzi and Clifton, 2015; Treiber et al., 2008), and air pollution emissions (Levy et al., 2010; Zhang et al., 2011). For air quality in particular, previous research has reported higher emissions during congested periods as vehicles spend more time idling, and engaging in more frequent acceleration and deceleration events (Smit et al., 2008; Zhang et al., 2011).

From this, increased air pollution emissions are associated with adverse human health effects (Cohen et al., 2017). A large number of epidemiological studies have shown that traffic emissions are linked to morbidity and mortality. These outcomes occur over multiple pathways with varying end points, which include respiratory and cardiovascular diseases (Peng et al., 2009; Phung et al., 2016), cancer (Fajersztajn et al., 2013; Guo et al., 2016), low birth weight (Bell et al., 2008; Veras et al., 2010), and diseases of the central nervous system (Genc et al., 2012; Suglia et al., 2008). For example, in Ontario, Canada, exposure to particulate matter 2.5 μg or smaller in aerodynamic diameter ($\text{PM}_{2.5}$) is associated with the development of diabetes, with a 1.11 hazard ratio for a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ (Chen et al., 2013). In China, Guo et al. (2016) show that the relative risks of lung cancer incidence related to a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ is 1.074 (95% CI: 1.060; 1.089). Pope et al. (2009) found that improvements in $\text{PM}_{2.5}$ exposure during 1980s and 1990s led to an average increase in life expectancy of 4.9 months in United States. In Connecticut and Massachusetts in the United States, $\text{PM}_{2.5}$ emitted by road transportation is associated with 2.1% and 3.5% increases in cardiovascular and respiratory admissions, respectively (Bell et al., 2014). In New York City, all on-road mobile sources contribute to about 320 deaths annually due to $\text{PM}_{2.5}$ exposures (Kheirbek et al., 2016).

$\text{PM}_{2.5}$ is a common air pollutant that has been associated with multiple adverse health outcomes. Numerous studies have shown that $\text{PM}_{2.5}$ is one of the major pollutants associated with traffic emissions. For example, motor-vehicle-related emissions are directly responsible for about

30% of ambient $PM_{2.5}$ in Toronto, Canada (Brook et al., 2007), 22% of $PM_{2.5}$ in Boston, U.S. (Masri et al., 2015), and 25% of $PM_{2.5}$ in five Chilean metropolitan regions (Kavouras et al., 2001). Worldwide, $PM_{2.5}$ is responsible for about 2.9 million deaths, according to the 2013 Global Burden of Disease Study (GBD, 2015). The World Health Organization (WHO) finds that $PM_{2.5}$ is responsible for an 8.6 month reduction in life expectancy for the average European population (WHO, 2013). In Canada, $PM_{2.5}$ was linked to 92,000 emergency department visits, and 21,000 early deaths in 2008 according to the Canadian Medical Association (CMA, 2008).

Considering this information, Zhang and Batterman (2013) argue that the public health impacts of traffic-related air pollution are the main economic externalities of congestion. However, while there is a wealth of previous research that has investigated the association between traffic emissions and human health, little work has employed measures of traffic congestion, including travel patterns, traffic volumes, and travel speeds, to directly account for the effects of congestion on air pollution. To our knowledge, Levy et al. (2010) and Zhang and Batterman (2013) were the only researchers that estimated health impacts based on traffic congestion, and their efforts were focused on congestion that occurs in the United States.

Considering that (i) congestion is a growing problem in many metropolitan areas worldwide; (ii) the literature shows strong evidence of the causal association of traffic emissions with morbidity and mortality, especially between $PM_{2.5}$ emitted by vehicles and mortality, and; (iii) there is a significant geographical gap of the research on attributable health risk in congestion periods; we argue that further investigation of the public health implications of traffic congestion is required.

Such work could enhance our understanding about the full costs of congestion, and how health risks differ by vehicle mix, fleet emissions characteristics, road infrastructure, population density, and atmospheric conditions. In response, in this present study we perform a health risk assessment of $PM_{2.5}$ emissions during congested periods in the Greater Toronto and Hamilton Area in Ontario, Canada. Specifically, we utilize a micro-level approach to estimate emissions based on estimated travel conditions. Next, we apply a concentration-response function to estimate $PM_{2.5}$ -related mortality and estimate the associated health costs.

2. BACKGROUND

2.1. Study area

The study was carried out in the Greater Toronto and Hamilton Area (GTHA), which is located in southern Ontario and the largest urban region in Canada. The GTHA encompass six health regions, including Hamilton, Halton, Peel, Toronto, York, and Durham (Figure 1). Health regions are administrative areas defined by provincial ministries of health according to provincial legislation.

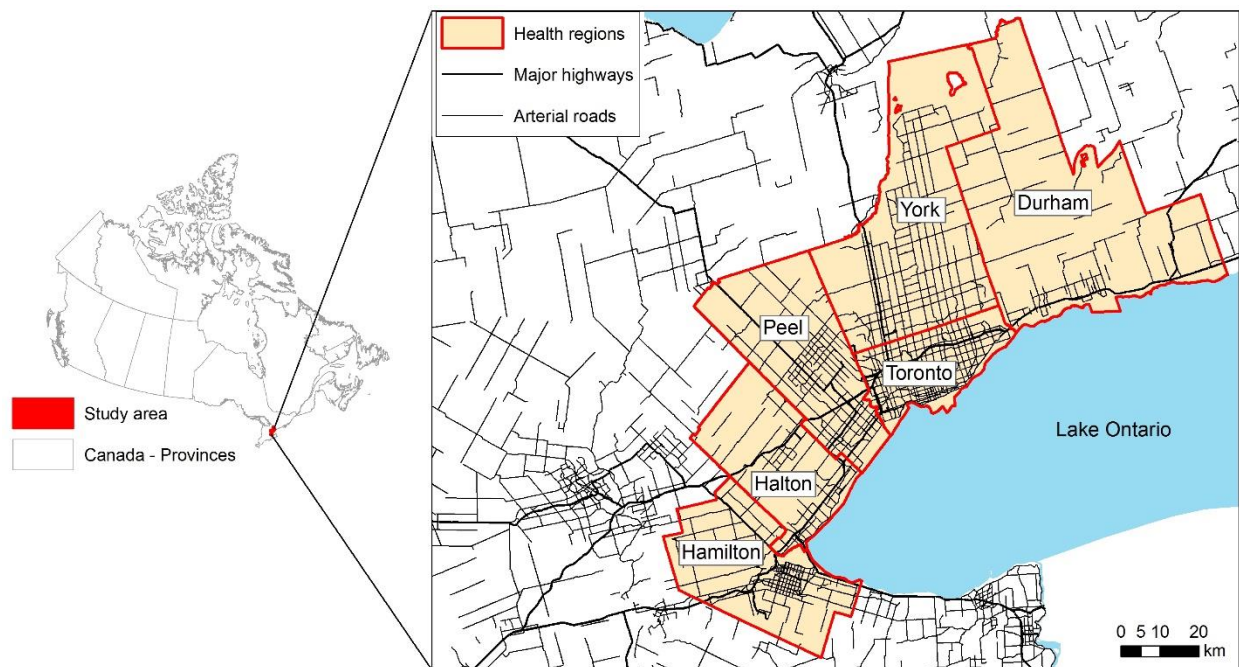


Figure 1 – Study area.

The population in the GTHA is estimated to be over 7 million in 2016, representing 17% of the Canadian population (SC, 2016). Traffic has been considered a critical concern in the GTHA for air pollution exposure (Adams et al., 2012; Adams and Kanaroglou, 2016). During 2009, within the province of Ontario, the Vehicle Kilometers Traveled (VKT) were about 130 billion, while within our study area (GTHA) VKT were approximately 55 billion (Natural Resources Canada, 2009). The indicator VKT is an aggregate measure of road usage expressed as the vehicle count on a given road link multiplied by length of the link. This indicator does not evaluate congestion. Details of how we incorporated congestion in our study are presented in section 3.1.

2.2. Traffic Congestion in the GTHA

Measures of the intensity of traffic congestion in the GTHA vary. Statistics Canada does not collect information on levels of congestion, but has started tracking commute durations as part of the 2011 National Household Survey. Here, it was estimated that the Toronto region has the highest commute times in Canada, with an average round-trip of 66 minutes compared to a national average of 51 minutes. Other data sources use proprietary methods to generate congestion indices, and generally offer a similar story. The 2016 TomTom Traffic Index for example places Toronto second behind Vancouver amongst Canadian cities for congestion with an estimated 30% increase in travel times on the road network compared to free flow conditions. The City of Hamilton is the least congested of the 12 cities in the TomTom data with a congestion index of 18%. Inrix, Inc. also calculates rates of traffic congestion for Canadian cities, and here, Toronto ranks second behind Montréal with an estimated 45.6 driver hours spent in congestion in 2016 measured against free flow travel conditions. In this data, Hamilton ranks ninth out of the 20 cities studied with 16.3 hours lost due to congestion. In terms of the geography of traffic congestion in the GTHA, previous research has used data on observed travel speeds (Sweet et al., 2015). This research examined differences in the spatial distribution of the extent and intensity of congestion and found that the highest levels of congestion are focused around Toronto's downtown core and suburban areas proximate to major highways.

3. DATA AND METHODS

This study was performed in five stages. In the first and second stages, we utilize traffic modeling techniques to infer traffic congestion and estimate vehicle emissions from daily travel in the GTHA. Next, we employ air pollution dispersion modeling to estimate exposure to congestion-related air pollutants. Fourth, we estimate a health risk using health assessment (concentration-response function), and the economic valuation associated with health impacts. Finally, in the fifth phase we performed a sensitivity analysis to assess the robustness of our results.

3.1. Traffic modeling

The first phase of the present research estimates link-level traffic conditions in the GTHA. To accomplish this, we employ the Stochastic User Equilibrium (SUE) traffic assignment algorithm to simulate how travelers choose their paths between origins and destinations in the

region on the road network. The particular origin and destination pairs for a given trip are defined *a priori* from a trip matrix derived from the 2011 Transportation Tomorrow Survey, a travel survey of more than 150,000 households in the GTHA (Data Management Group, 2013). The trip matrix represents motorized passenger trips that occurred on a typical weekday throughout pre-defined, mutually exclusive traffic analysis zones in the study area. Note that this matrix does not include commercial vehicle movements or transit trips.

The algorithm SUE estimates congested travel times for each link using the congestion function proposed by the US Bureau of Public Roads (US Bureau of Public Roads, 1964). This function utilizes link design capacity (flows, measures in vehicle units) to assign trips while considering link congestion levels (traffic volume to capacity ratio). In this sense, the congestion captured by this approach is a function of estimated link flows and capacity. Such inferred congestion differs from congestion that might be observed through the travel speed data in the GTHA discussed above. Our inferred measures of congestion are also ‘predictable’ in the sense that we capture what are assumed to be daily increases in travel time associated with link capacity. While such predictable daily congestion impacts drivers, research has shown that the unpredictability of congestion, which can result in unreliable travel times, is a major source of stress among drivers (Wener and Evans, 2011). In this sense, our measures of traffic flows capture only an element of traffic congestion. Nevertheless, despite these limitations, our simulation of traffic conditions provides a foundation on which estimates of emissions can be developed. Estimates of weekday passenger vehicle flows were performed as follows:

$$t(a, n) = t^f(a) \left(1 + \alpha \left(\frac{flows(p, a, n)}{capacity(a)} \right)^\beta \right) \quad \text{Equation 1}$$

where, $t(a, n)$ is the flow travel times on link a ; n is an index referring to the iteration number when convergence is achieved. In other words, traffic assignment submodels are repeated until all inter-zonal congested travel times stabilize. The result here represents an inter-zonal travel time matrix corresponding to a congested situation; $t^f(a)$ is free flow travel times on link a ; α and β are constants set to 0.15 and 4, respectively; $capacity(a)$ is the design capacity of link a ; and $flows(p, a, n)$ is the flow of light duty passenger vehicles on link a in iteration n .

Once flows are assigned to the network, congested speeds can be calculated for each link, and this information was employed to estimate emissions from traffic flows. For each link the output is an average congested speed and an estimated traffic volume. In other words, the model will result in lower overall speeds when the number of trips is high relative to capacity. Figure 2 shows the average VKT and average speed by hour, illustrating a snapshot of congestion as result of the traffic modeling.

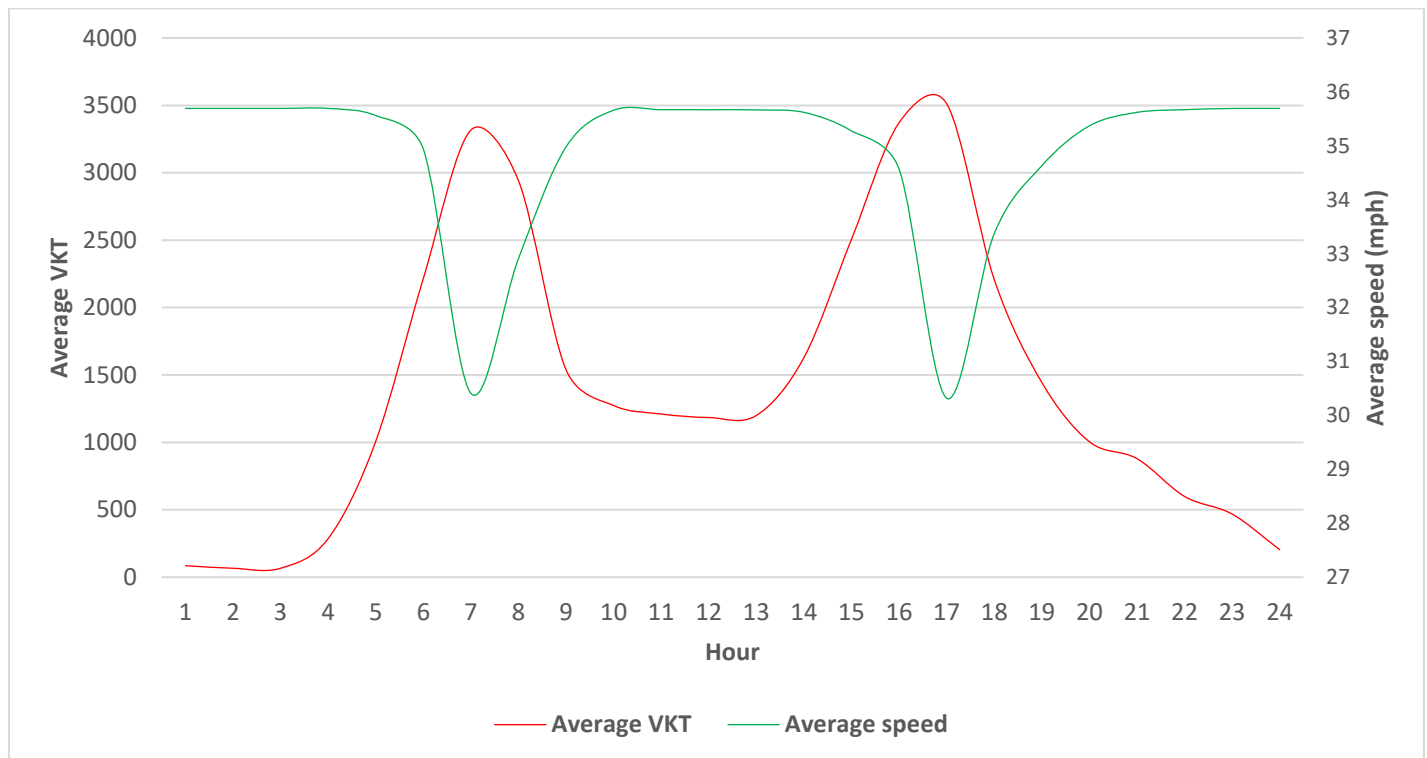


Figure 2 – Average VKT and average speed by hour in the GTHA.

3.2. Emissions modeling

We used US EPA’s MOVES model to estimate PM_{2.5} emissions in grams per kilometer (exhaust + brake + tire emissions) on an hourly basis, for the year 2011 based on the traffic assignment data. An overview of the process is presented below, and the full details are presented in Requia et al. (2017).

We considered only passenger vehicles emissions in this study. In generating these emissions, MOVES relies on numerous data inputs to reflect the environmental conditions

(temperature and relative humidity), fuel and vehicle fleet characteristics (vehicle population, age distribution, fuel input), and vehicle activity of study area (speed distribution).

Temperature and relative humidity were based on hourly/monthly historical averages from 1980 to 2010, and obtained from Environment Canada's historical climate data portal (<http://www.climate.weather.gc.ca>). Data on fuel and vehicle fleet characteristics were obtained from Polk Canada (<https://www.ihsc.com>). Road link speed distributions of vehicular traffic were obtained from INRIX Corporation (<http://www.inrix.com>). This dataset reports the average speed on specific road segments in the GTHA in 15-minute intervals. We processed this data to derive the 24-hour average speed distribution for four road types (rural restricted access, rural unrestricted access, urban restricted access, and urban unrestricted access) by weekday. Hourly emissions on the street network were generated by integrating the emission factors and traffic flows. The street network and associated traffic flows and emissions were combined within a Geographic Information System (GIS) framework to map the results.

To emphasize the effect of low speeds on the generation of high emissions, Figure 3 shows the running emission rates for PM_{2.5} (exhaust, brakewear, and tirewear) measured in grams per kilometer. There is a general theme in Figure 2 that the lowest of speeds are associated with much higher emission rates. On the other hand, there are consistently low emission rates associated with higher average speeds (> 45 mph). Higher speeds have fewer periods of idling and less acceleration from idling, while low speeds feature much more of such vehicle activity.

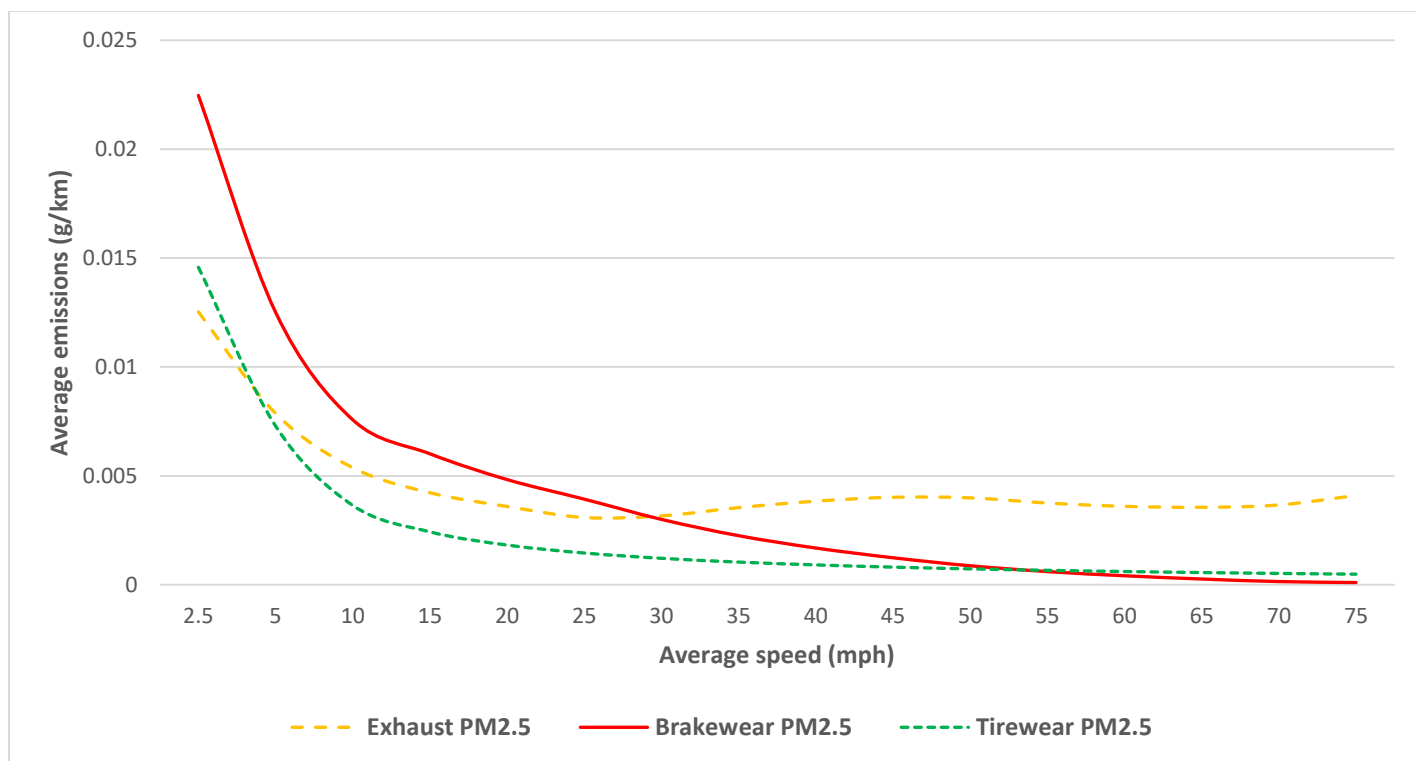


Figure 3 – Running emissions rates by speed estimated in our analysis.

3.3. Exposure modeling (dispersion modeling)

Using the emission values, we determined the increase to ambient $PM_{2.5}$ (annual average per hour) across the GTHA using the R-line model, version 1.2. This model is designed for near-roadway assessments, and it is currently under development by the Environmental Protection Agency (EPA). The R-line model is a grade dispersion approach based upon a steady-state Gaussian formulation by numerically integrating point source emissions. Information on the model formulation can be found in (Snyder et al., 2013).

In our analysis model inputs included traffic emissions, receptor locations, and surface meteorological data. Receptors were located at centroid of a 1 km x 1 km grid over the GTHA. The meteorological input data was provided by the Ontario Ministry of the Environment (OME, 2009). This data contains hourly information on surface friction velocity, convective velocity scale, Monin-Obukhov length, surface roughness height, wind speed, and wind direction, which are derived from surface characteristics, cloud cover, upper air temperature soundings, near surface wind speed, wind direction and temperature data.

3.4. Health Assessment and Economic Valuation

We assessed health risks (total non-external mortality and cardiovascular mortality) by linking estimated exposures to the concentration-response relationships from the literature. The concentration-responses function was based on the published cohort study from Farhat et al. (2013). Their study report relative mortality risk (RR) per 10 $\mu\text{g}/\text{m}^3$ increase in 24-hours average $\text{PM}_{2.5}$ (short-term exposure, acute effects) in 12 Canadian Cities, including two cities located in the GTHA, Hamilton and Toronto. In our analysis we used the pooled estimates across Hamilton and Toronto. For all-cause mortality, the RR was 1.06 [1.02 – 1.11], and for cardiovascular mortality the RR was 1.12 [1.08 – 1.15].

Mortality data of each health region (base number of annual health outcome) were used in the health assessment modeling. These data were provided by Statistics Canada (<http://www12.statcan.gc.ca/health-sante>). In appendix 1, we present the health data by region.

We estimated mortality outcome for two temporal scales: day (Equation 2) and hour (Equation 3).

$$H_d = \frac{\sum_{h=1}^{24} C_{dh}}{24} \times B \times \beta \times \left(\frac{5}{7}\right) \quad \text{Equation 2}$$

$$H_{dh} = \frac{C_{dh}}{\sum_{h=1}^{24} C_{dh}} \times H_d \quad \text{Equation 3}$$

Where H_d is daily health outcome for grid d ; C_{dh} is $\text{PM}_{2.5}$ concentration ($\mu\text{g}/\text{m}^3$) for grid d for hour h ; B is base number of health outcome; β is percentage change in health outcome per unit increase of $\text{PM}_{2.5}$; and, H_{dh} is health outcome associated to grid d and hour h . It is important to note that Equation 2 calculates the total health outcome associated with the daily average increase to ambient air pollution concentrations from congestion, Equation 3 determines the effect associated to the emissions of each hour of the day, and (5/7) is the adjusted coefficient in order to account for exposure during weekdays.

For the economic valuation of these health effects, we used the Value of Statistical Life (VSL) to estimate the cost of mortality attributable to congestion. According to the Policy Horizons Canada – A Canadian organization within the federal public service

(<http://www.horizons.gc.ca/eng>), the central estimate of VSL for Canadian policy analysis is \$6.5 million (2007 Canadian dollars), with lower bound equal to \$3.5 million and higher bound equal to \$9.5 million.

Finally, we highlight that PM_{2.5} was the focus of this study for risk assessment because of the following reasons: (i) there is a strong scientific evidence showing that PM_{2.5} presents a causal association with health outcomes (Schwartz et al. 2015; Schwartz, Bind, and Koutrakis 2016), (ii) we used R-line model, which it is indicated for evaluation of air quality impacts in the near-road environment. The R-line model estimates only particulate concentrations, and, (iii) we performed the health assessment considering short-term exposure in order to estimate mortality outcome for two temporal scales – day and hour. Here, again, the epidemiological studies have established robust causal associations between short-term exposure to PM_{2.5} and mortality.

3.5. Sensitivity analysis

We conducted sensitivity analyses in order to examine the robustness of the risk assessment and the economic valuation. Sensitivity analysis was performed using Monte Carlo simulation, which is common in risk assessment (Ji et al., 2015, 2012; Pascal et al., 2013; Xu et al., 2015). We assessed 10,000 simulations with a 95% confidence interval (normal distribution) to simulate the parameters, and triangular distribution for each input variable. The range considered for the generation of random values for each variable is presented below.

For the health assessment, we tested the sensitivity of three variables: the base number of annual health outcomes measured by mortality, relative risk from literature (concentration-response), and PM_{2.5} concentrations. The range of random values of the mortality data varied between (-5%) and (+5%) of the base case. This is the range indicated by Statistics Canada (<http://www12.statcan.gc.ca/health-sante>). For the relative risk, we considered a range from the lower bound to the upper bound of the confidence level (95%), according to the literature considered (Pope and colleagues). Finally, for PM_{2.5} concentration, the range of random values was from the lower bound to the upper bound of the confidence level (95%) of the daily mean.

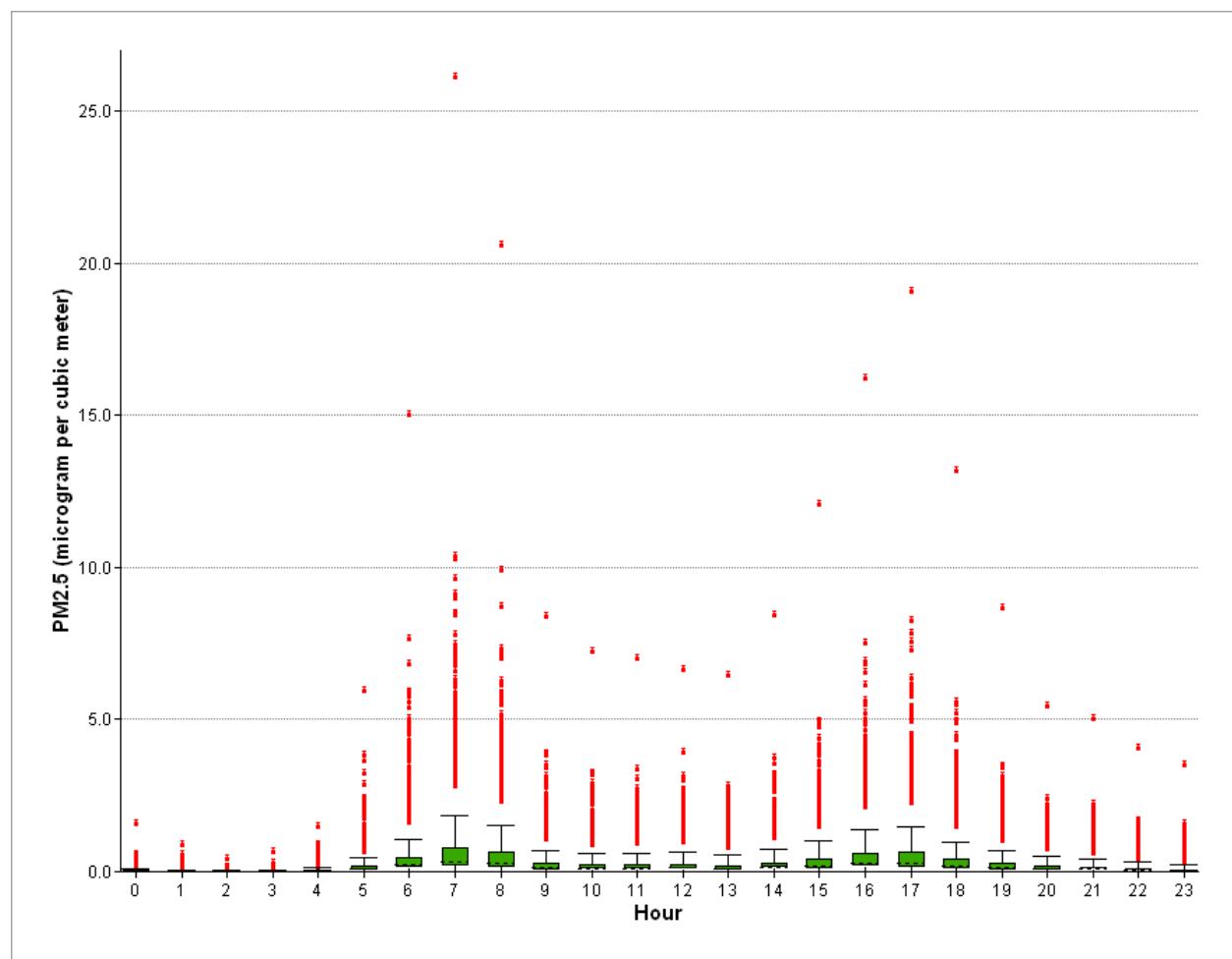
In the sensitivity analysis for the economic valuation we considered random values for two variables: PM_{2.5}-related mortality risks (derived from our health assessment) and the VSL. For PM_{2.5}-related mortality, we defined the range of random values according to the confidence level

(95%) estimated in the previous sensitivity analysis (health assessment). For the VSL, the range values were based on the lower and upper bounds of \$3.5 million and \$9.5 million, respectively.

4. RESULTS

4.1. Spatiotemporal patterns of PM_{2.5} concentrations

We estimated an average increase to ambient PM_{2.5} 24-hour concentrations due to passenger vehicle congestion of 0.24 (SD 0.32) $\mu\text{g}/\text{m}^3$. The low average value occurs because a large number of grid cells are not affected by transportation emissions. Comparing this value with the baseline PM_{2.5} concentration, in 2016, government monitoring identified the mean PM_{2.5} concentration was 4.8 $\mu\text{g}/\text{m}^3$, which was measured at the Toronto Downtown Air Pollution Monitor operated by the Ontario Ministry of the Environment and Climate Change. The grid cell with the highest 24-hour average increase of 8.54 $\mu\text{g}/\text{m}^3$ is located in Toronto, which has the highest hourly concentration increases. Observing the average emissions by hour over all health regions, the highest exposures occur during two periods, between 7:00 am (mean 0.67, SD 0.90 $\mu\text{g}/\text{m}^3$) and 8:00 am (mean 0.55, SD 0.75 $\mu\text{g}/\text{m}^3$), and between 4:00 pm (mean 0.49, SD 0.62 $\mu\text{g}/\text{m}^3$) and 5:00 pm (mean 0.54, SD 0.71 $\mu\text{g}/\text{m}^3$). These are the most congested periods during the day, which account for the highest emissions of PM_{2.5}. Particularly at 7:00 am, we observed a grid cell located in Toronto with the highest PM_{2.5} concentration increase during the day of 26.20 $\mu\text{g}/\text{m}^3$. We present in Figure 4 the PM_{2.5} increases due to passenger vehicle congestion by hour in the region, illustrating the Interquartile Range (IQR) and outliers from each period. Figure 5 includes the spatio-temporal distribution of PM_{2.5} increases due to passenger vehicle (a video showing this spatio-temporal distribution can be found online as Supplemental Material). In appendix 2, we present average PM_{2.5} concentration by health region and hour.



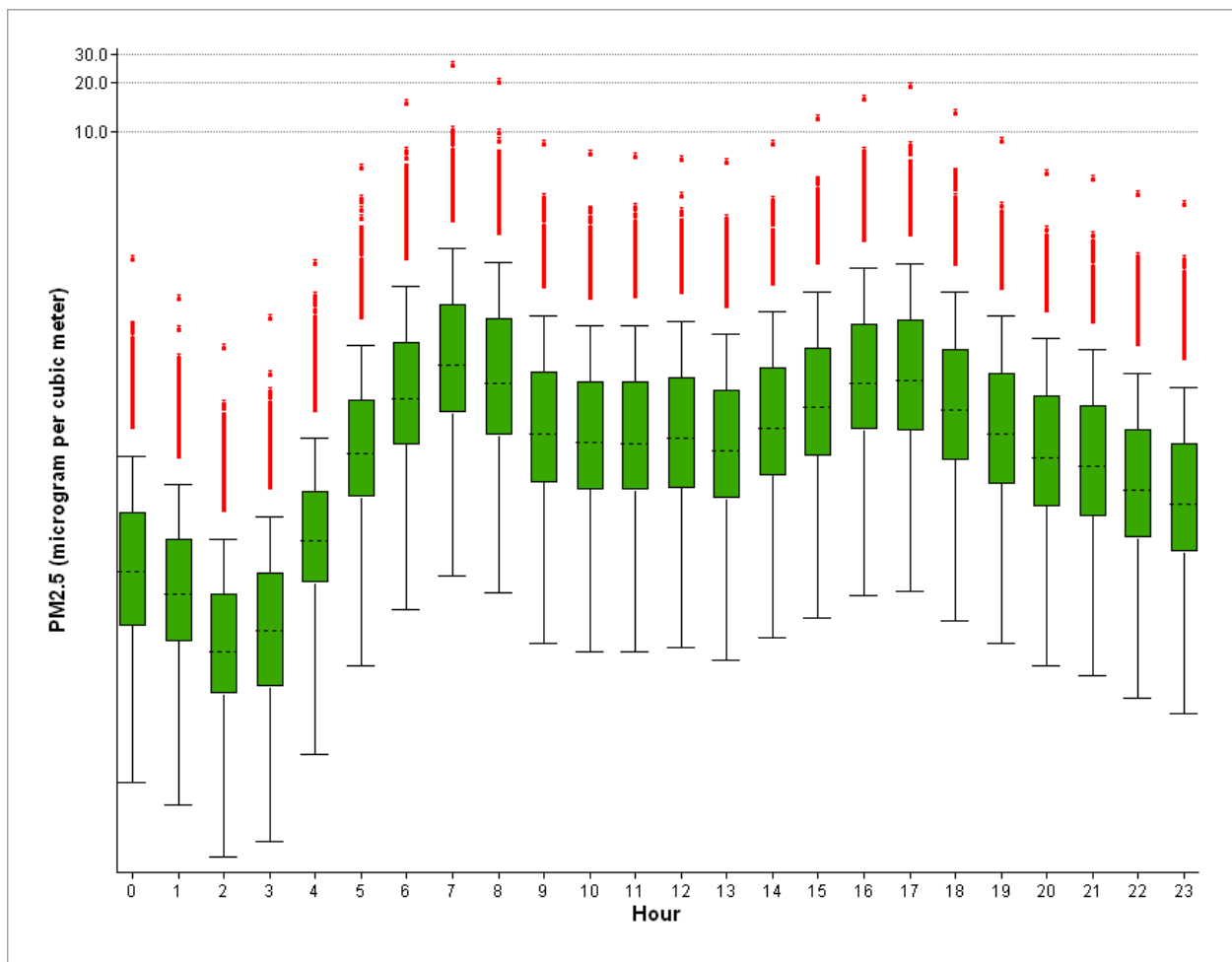


Figure 4 – PM_{2.5} increases due to passenger vehicle congestion by hour in the GTHA. Note: chart on the top (actual data); chart on the bottom (log scale is used for the Y axis for a better visualization of the distribution); Interquartile range, IQR (green); outliers (red).

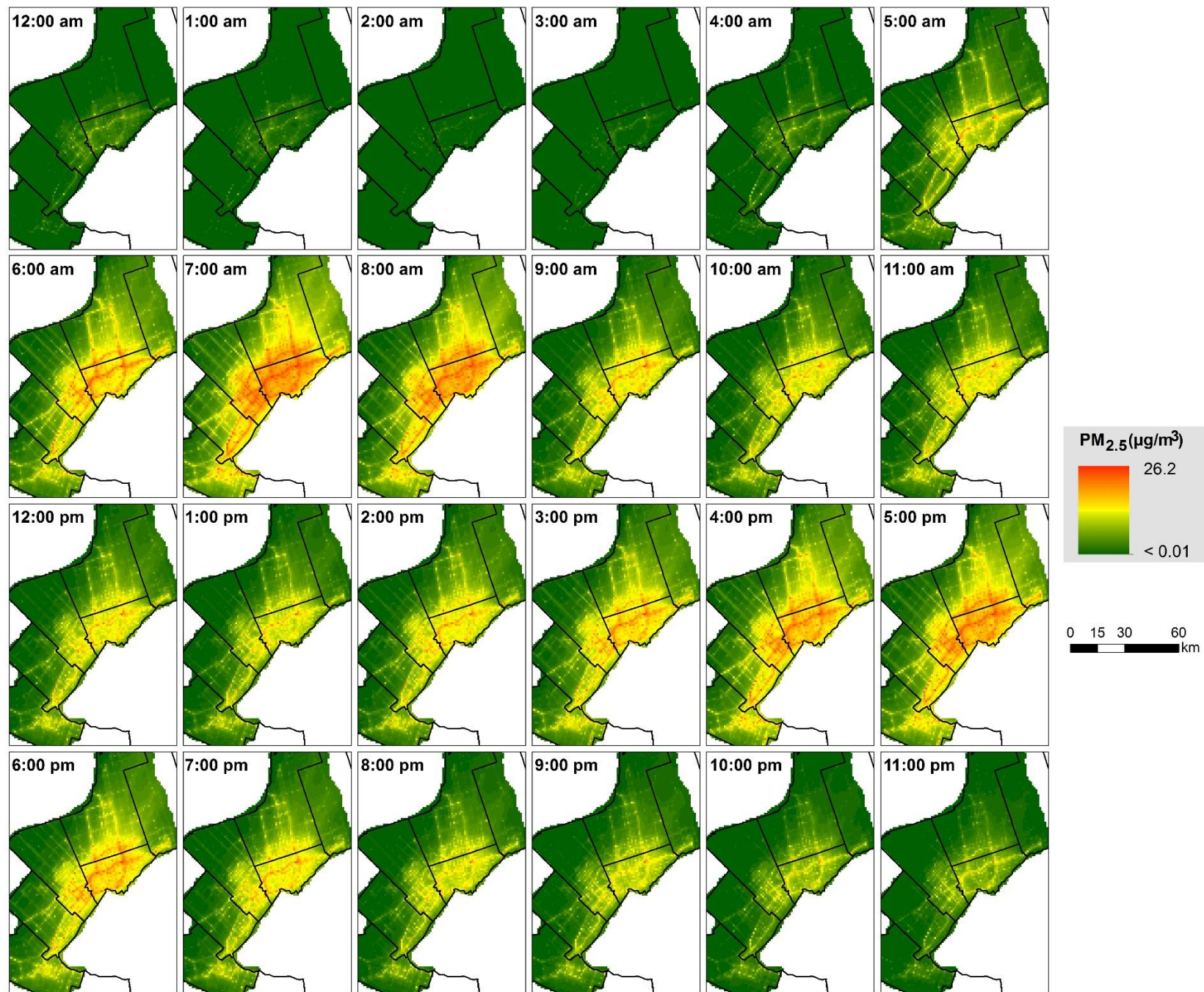


Figure 5 – Spatio-temporal distribution of $PM_{2.5}$ increases due to passenger vehicle congestion in the GTHA.

4.2. Health impacts and its monetized value

On average, Toronto accounts for approximately 77% of the total health impact in the GTHA. Durham, Halton, Hamilton, Peel and York Regions account for about 3, 4, 3, 8, and 5% respectively (Appendices 3 and 4). Figure 6 shows the PM_{2.5}-related mortality risks (all-cause and cardiovascular) and its monetary valuation in the GTHA (considering all health regions). In appendices 3 and 4 we present the mortality risks by health region and hour, and in appendix 5 and 6 we present the total health impacts considering the 24 hour average.

We estimate that the highest health impacts of PM_{2.5} emissions are attributable to traffic congestion from 7:00 am – 7:59 am in the GTHA. The annual all-cause mortality during this period was approximately 23 deaths, and for annual cardiovascular mortality was 13 deaths (Figure 6). The proportion of the health impacts attributable to congestion varies substantially across the time of day. For example, on average the annual mortality (all-cause mortality and cardiovascular mortality over the GTHA) predicted for 7:00 am (most congested period) is about 59 times higher than for 2:00 am, 44 times higher than for 3:00 am, and 18 times higher than for 4:00 am.

The public health cost associated with weekday traffic congestion at the most critical period (7:00 am) in the GTHA (all health regions) was approximately \$151 million dollars per year for all-cause mortality, and \$88 million dollars for cardiovascular mortality (Figure 6). Considering daily mortality in the GTHA, we estimate this cost to be \$1.3 billion and \$778 million for all-cause and cardiovascular mortality, respectively. In appendices 7 and 8 we present the monetary valuation by health region and hour.

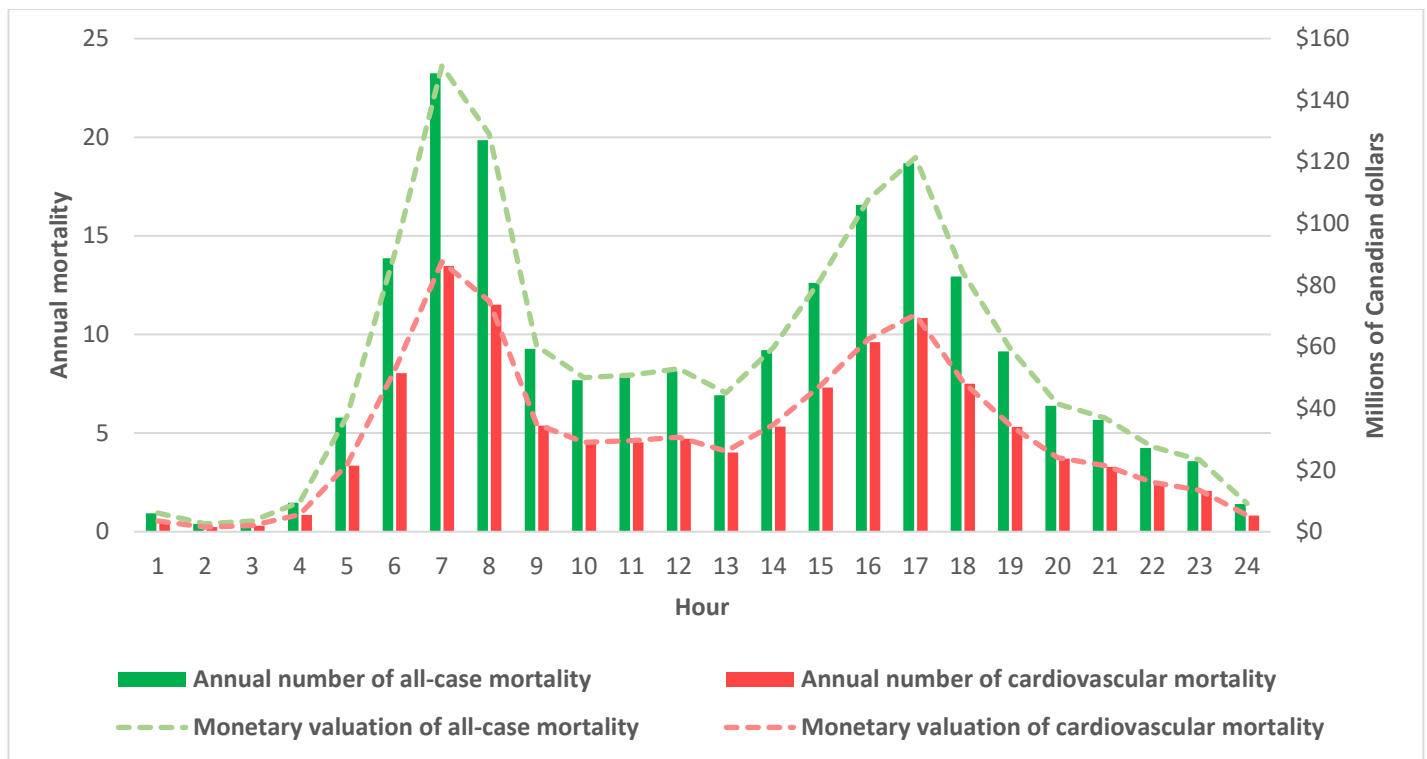


Figure 6 – PM_{2.5}-related mortality risks (all-cause and cardiovascular) and its monetary valuation in the GTHA.

4.3. Sensitivity analysis

Results of the Monte Carlo analysis are presented in Tables 1 and 2, and in Figures 7 and 8. The sensitivity analysis presented here is based specifically on traffic congestion at 7:00 am in Toronto. Since that period and that health region are the most affected by traffic congestion, the sensitivity analysis results shown here can provide a sense of the likelihood of significantly different conclusions.

In general, our results are relatively sensitive. For example, in the primary analysis we observed a total of 18 all-cause mortalities attributable to PM_{2.5} in Toronto at 7:00 am (Appendix 3), while in the sensitivity analysis the estimated average all-cause mortality was 18.7 (Table 1). The boundary test for this example from Table 1 shows a lower and upper bound equal to 9.7 and 28.1, respectively. This boundary test represents the probability (95%) of the total value (in this case, mortality risk). Figures 7 and 8 show the overall view of the distribution between the minimum and maximum values of the mortality risk and monetary valuation. The boundaries with 95% probability are highlighted in these Figures.

Table 1 – Sensitivity analysis results (summary statistics) of the risk assessment and its monetized value.

Statistical parameter	Mortality risks		Monetary valuation (Millions of C\$)	
	All-cause	Cardiovascular	All-cause	Cardiovascular
Minimum	5.4	6.2	40.5	30.3
Maximum	35.1	14.8	254.4	110.1
Average	18.7	9.9	120.7	65.3
Standard deviation	5.5	1.4	33.5	13.8
Boundary test (95%)	9.7 ; 28.1	7.6 ; 12.2	70.8 ; 180.2	43.2 ; 89.1

Note 1: Boundary test (95%) represents the probability of the total value (mortality risks, monetary valuation) between the lower and upper bound. We considered here 95% of probability.

Note 2: The results presented in this table are based on the analysis of Toronto (health region) and on the period of 7:00 am.

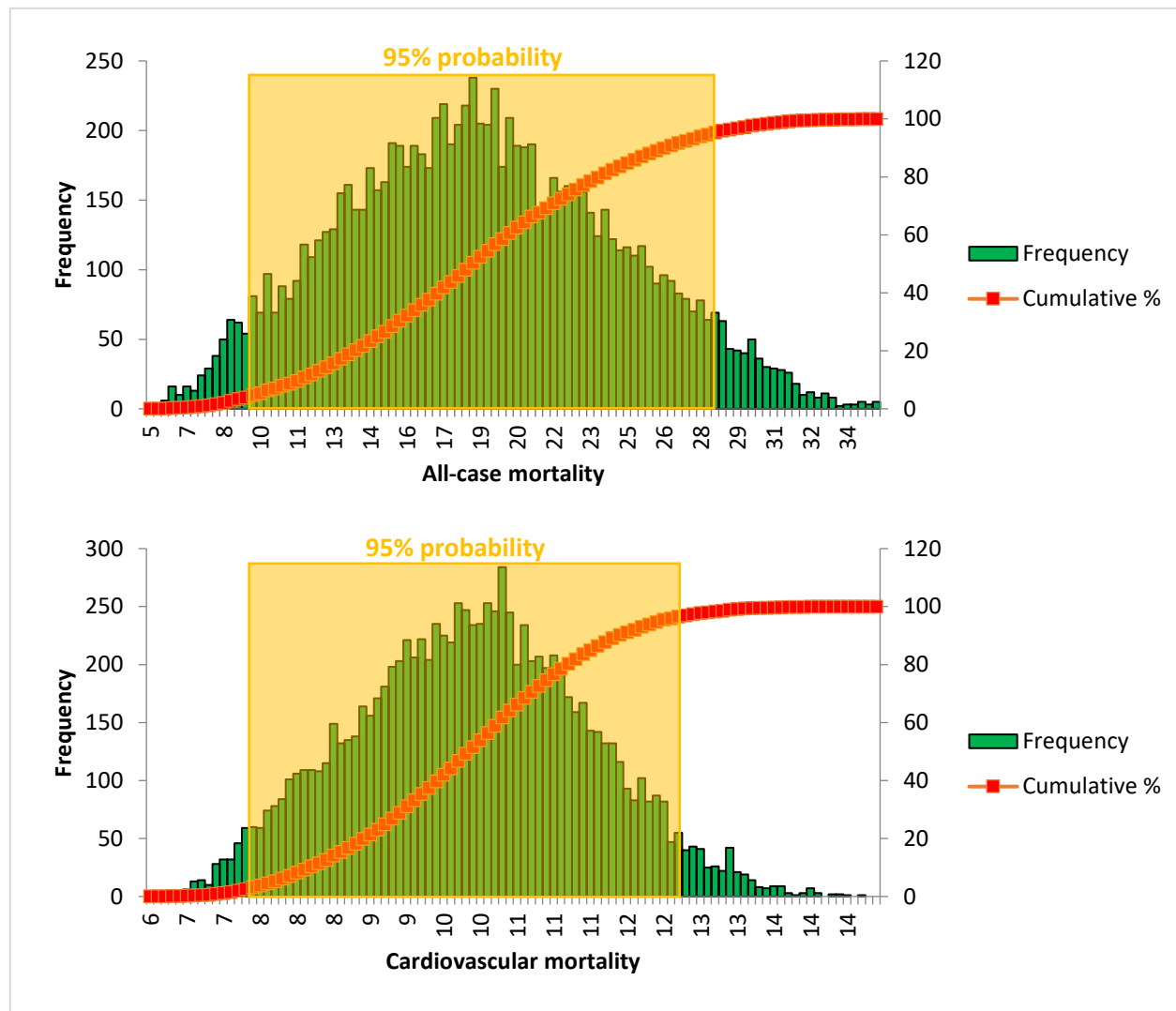


Figure 7 – Sensitivity analysis – frequency and cumulative percentage of the all-cause (top) and cardiovascular mortality (bottom).

Note: The results presented in these charts are based on the analysis of Toronto (health region) and on the period of 7:00 am.

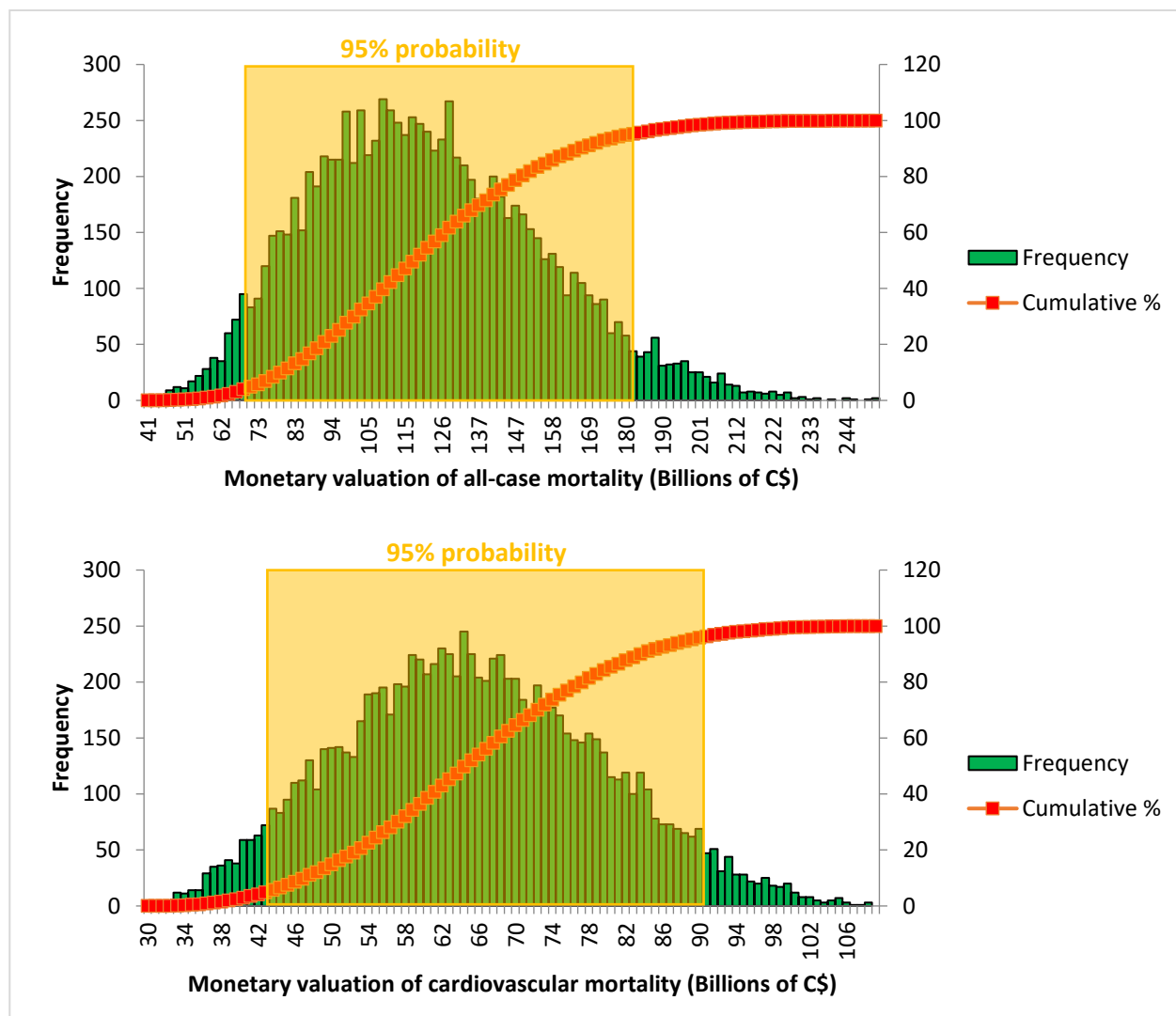


Figure 8 – Sensitivity analysis – frequency and cumulative percentage of the monetary valuation of all-cause (top) and cardiovascular mortality (bottom).

Note: The results presented in these charts are based on the analysis of Toronto (health region) and on the period of 7:00 am.

When we examine the correlation coefficients from the Monte Carlo simulation (Table 2) for the health assessment (mortality risk), relative risk (representing the concentration response) is the most influential variable in our analysis. The correlation coefficient is equal to 0.96 and 0.94 for all-cause and cardiovascular mortality, respectively. For the monetary valuation, PM_{2.5}-related mortality risk was the variable with the highest correlation coefficient, with a value of 0.73 for all-cause mortality and 0.89 for cardiovascular mortality. The closer the correlation coefficient is to 1, the greater the influence of the variable on the outcome (mortality risk or monetary valuation value).

Table 2 – Sensitivity analysis – correlation coefficient for the risk assessment and its monetized value.

Variable	Mortality risks		Monetary valuation (CAD\$)	
	All-cause	Cardiovascular	All-cause	Cardiovascular
Relative risk from literature (concentration-response)	0.96	0.94	-	-
PM _{2.5} concentration	0.11	0.33	-	-
Base number of annual health outcome (mortality data)	0.05	0.14	-	-
PM _{2.5} -related mortality risks (derived from our health assessment)	-	-	0.73	0.89
VSL	-	-	0.65	0.43

Note: The results presented in this table are based on the analysis of Toronto (health region) and on the period of 7:00 am.

5. DISCUSSION AND CONCLUSIONS

Our study suggests that weekday traffic congestion has a substantial impact on human health and the economy, especially during the morning peak period (7:00 am – 7:59 am). Health and economic impacts generated during this period can be nearly 59 times higher than the least congested period (2:00 am). Considering only the traffic emissions at 7:00 am, we estimate that PM_{2.5}-related all-cause mortality in the GTHA is approximately 23 deaths per year (boundary test 95%: 12; 34), and cardiovascular mortality is about 13 deaths per year (boundary test 95%: 7; 17). The monetized value of these health impacts is approximately \$151 million (boundary test 95%: \$79 million; \$211 million) due to all-cause mortality, and \$88 million for cardiovascular mortality (boundary test 95%: \$49 million; \$122 million). Considering the daily average concentrations, our results indicate an effect of 206 (boundary test 95%: 116; 297) and 119 (boundary test 95%: 67; 171) deaths per year for all-cause and cardiovascular mortality, respectively. The economic impact from daily mortality is approximately \$1.3 billion (boundary test 95%: 0.8; 1.9), and \$778 million (boundary test 95%: 478; 981), for all-cause and cardiovascular mortality, respectively.

Our estimates are quite similar to those previously reported for specific regions in Canada. For example, in Toronto, the local Public Health Agency shows that traffic-related air pollution is linked to about 440 premature deaths per year. According to this report, mortality-related costs associated with traffic pollution in Toronto are about \$2.2 billion (Toronto Public Health, 2007). In our study, Toronto accounted for approximately 77% of the total health impact in the GTHA, which was the highest of any health region. Our results show that in Toronto, daily mortality congestion-related air pollution can cause about 161 premature deaths per year, with a health cost

equal to \$1 billion. The methods used in our study may explain the difference between our results and those reported by the Toronto Public Health (2007). For example, while we considered a micro-level approach to estimate emissions based on a snapshot of congestion and focused only on PM_{2.5} emissions related to passenger cars, the public health agency in Toronto used a basic model to estimate traffic emissions (traffic flow x vehicle classification volumes x emission factors) with no congestion analysis, and the agency accounted for all type of vehicles and a mix of pollutants, including CO, NO₂, SO₂, O₃, and PM.

Other local agencies have reported results considering only exposure to PM_{2.5} ambient concentration (no traffic analysis) for particular areas in the GTHA. For example, in Hamilton, PM_{2.5} ambient concentration is linked to 40 premature deaths annually (Clean Air Hamilton, 2012). In our study, specifically in Hamilton, we estimated 7 premature deaths per year (daily mortality, all-cause). In neighboring Waterloo region, PM_{2.5} ambient concentration accounted for an estimated 127 non-traumatic deaths, chronic exposure (Health, 2008).

Provincial and national reports have also shown the harmful effects of air pollution on health, which are parallel to our findings. The Ontario Medical Association (OMA, 2005) estimated that in 2005, overall economic losses associated with air pollution exposure (ambient air pollution) were expected to be approximately \$7.8 billion in Ontario. Our results show that only in the GTHA, the public health cost of daily PM_{2.5}-related traffic congestion is about \$1.3 billion (boundary test 95%: 0.8; 1.9) for all-cause mortality. From a national perspective, the economic cost of mortality due to PM_{2.5} and O₃ was estimated to be between \$27 and \$49 billion in Canada in 2015 (Brauer et al., 2016; Forouzanfar et al., 2015). This cost is linked to approximately 7,712 deaths attributable to PM_{2.5} and O₃ ambient concentration. PM_{2.5} accounted for by far the largest share of these deaths. In our research, considering daily exposure, we observed a total of 206 (boundary test 95%: 116; 297) and 119 (boundary test 95%: 67; 171) deaths per year (all-cause and cardiovascular mortality, respectively).

Our findings are also consistent with the limited existing studies in the international literature. Levy et al. (2010) estimated PM_{2.5}-related premature mortality attributable to traffic congestion in 83 urban areas in the U.S. Their estimates shows that PM_{2.5} emitted by traffic congestion caused about 1,100 premature deaths in these 86 cities in 2010 with a cost estimated in 18 billion dollars. Only in Boston, Chicago, and Los Angeles, Levy et al. (2010) show that the

monetized estimates of the PM_{2.5}-related mortality risks from traffic congestion is 100 million, 600 million, and 1.5 billion dollars, respectively.

Zhang et al. (2013) considered NO₂ emissions to estimate the relationship between congestion and health risks in two selected roads in Detroit, U.S. They indicate that congestion during the morning rush hour may increase the risks by 20 to 40% compared to afternoon rush hour. In our study we observed that congestion at 7:00 am increased the risks by 28 to 42% compared to 4:00 pm. We highlight that to the best of our knowledge, Levy et al. (2010) and Zhang et al. (2013) were the only researchers that estimated human exposure based on congestion conditions.

Our findings include some uncertainties. First, we only accounted for congestion impacts associated with increases in ambient concentration. We did not consider other potential environment of exposure due to emissions during congestion, including in-cabin exposure (exposures for drivers and passengers) and indoor exposure (homes and workplaces). We also did not account for exposure during weekends and emissions from other mobile sources, including buses, trucks, and motorcycles. Since we only considered passenger cars the health burden may be underestimated. However, according to Statistics Canada, passenger vehicles represent about 94% of the total vehicle fleet, while trucks, buses, and motorcycles represent 3, 1, and 2%, respectively. Future research accounting for these attributes is highly recommended. Third, we did not validate PM_{2.5} concentrations, since our estimates represent only PM_{2.5}-related traffic congestion; source apportionment in future research could be used for validating the contribution to ambient PM from transportation sources. Fourth, we did not estimate other public health and economic impacts which may occur during congestion periods, including time and fuel wasted, which according to Levy et al. (2010), are significant enough in magnitude to be considered in strategies to mitigate congestion. We also only assessed mortality risk due to PM_{2.5} emissions. The literature has shown that many other air pollutants (e.g., NO₂, O₃) and health outcomes (e.g., respiratory diseases, hospital admissions) may represent other health impacts from congestion.

Finally, the framework implemented in our analysis can lead to errors in estimates of vehicular emissions based on congestion conditions. For example, rate per vehicle emissions, which are not related to distance travelled, have been assigned to links via a proportional allocation process. As consequence, it is possible that congestion has been under or over-estimated relative to reality. Clearly, it is very difficult to take daily traffic variation into account.

However, despite the limitations, our study can be an important tool in providing guidance to public policymakers. Previous studies have accounted for congestion considering aggregate indicators, and regarding the spatial scale, previous investigations have considered urban area as a whole or only small sample of roads. In comparison, we used a modeling approach (micro-simulation) that detects spatial differences at fine scale in terms of roads conditions and traffic flow. Finally, we estimated health and economic impacts considering daily exposure and were able to assign the health outcomes by hourly variation. Previous epidemiological studies have shown that hourly peak PM_{2.5} may be an important risk factor of mortality (Lin et al., 2017). According to this recent study, there is a greater mortality burden using hourly peak PM_{2.5} than daily mean PM_{2.5}.

This study expands the scientific knowledge on the impacts of traffic congestion. Better knowledge of this relationship should be of interest to policy makers to create future strategies related to environmental health and transportation infrastructure. In addition, taken together, results from this study will better prepare health experts and environmental managers for the challenges of regulating land use, transportation, and air quality in a changing effect. Reliable projections of transportation and air pollution levels will improve the capability of the medical community to prepare for future trends.

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